

“Battery Management System for Microgrid”

*Project report submitted to
Visvesvaraya National Institute of Technology, Nagpur
in partial fulfilment of the requirements for the award of
the degree.*

Bachelor of Technology In Electrical and Electronics Engineering

by

**M. Ratna Persis
(BT20EEE063)**

**Y. Abhinav
(BT20EEE112)**

**Y. Harshal Sunil Rao
(BT20EEE115)**

under the guidance of

Dr. Avisha Tah



**Department of Electrical and Electronics
Visvesvaraya National Institute of Technology
Nagpur 440 010 (India)**

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Declaration

We, **Mallepogu Ratna Persis (BT20EEE063)**, **Abhinav Yellamelli (BT20EEE112)**, **Harshal Sunilrao Yerne (BT20EEE115)** hereby declare that this project work titled **“Battery Management System for Microgrid”** is carried out by us in the Department of Electrical Engineering of Visvesvaraya National Institute of Technology, Nagpur. The work is original and has not been submitted earlier whole or in part for the award of any degree/diploma at this or any other Institution / University.

Date:

Sr.No.	Enrollment No	Names	Signature
1.	BT20EEE063	Mallepogu Ratna Persis	
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3.	BT20EEE115	Harshal Sunilrao Yerne	

Certificate

This is to certify that the project titled “Battery Management System for Microgrid”, submitted by **Mallepogu Ratna Persis (BT20EEE063), Abhinav Yellamelli (BT20EEE112), Harshal Sunilrao Yerne (BT20EEE115)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electrical and Electronics Engineering**, VNIT Nagpur. The work is comprehensive, complete and fit for final evaluation.

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Assistant Professor, Electrical
Engineering, VNIT, Nagpur

Professor B.S. Umre,
Department of Electrical Engineering
VNIT, Nagpur
Date:

Acknowledgement

We would like to express our gratitude to our respected guide, Dr. (Ms) Avisha Tah, whose insightful guidance and knowledge was paramount in aiding us to carry out this project successfully. We are extremely grateful for her continuous support and presence whenever we needed it. We would also like to acknowledge our respected panel members for their constructive and insightful remarks which helped us identify and improve on our weaker areas.

Engaged in this project, we have delved into an innovative domain with expansive potential for advancement. Our endeavors encompassed a wide array of facets within this revolutionary field, affording us an extensive exploration. Throughout this journey, we acquired indispensable skills and effectively translated theoretical understanding into tangible outcomes. The experience proved profoundly motivational and enlightening, fostering our growth and proficiency. Gratitude is extended to all contributors, both directly and indirectly involved in the project. Additionally, it is imperative to acknowledge the collaborative efforts that facilitated our progress and enriched our learning.

Abstract

This project focuses on the development of a Battery Management System (BMS) tailored for microgrid applications. Battery Management Systems (BMS) serve as indispensable components in ensuring the dependable operation and effective utilization of battery energy storage systems (BESS) within microgrid environments. This abstract provides an overview of the significance, functionalities, challenges, and advancements in BMS technology tailored specifically for microgrid deployments.

Within the microgrid context, BMS functionalities extend beyond conventional battery monitoring and protection to encompass grid integration, load management, renewable energy optimization, State of Charge (SoC) estimation, and cell balancing techniques. The BMS facilitates seamless interaction between battery storage, renewable energy sources, and the main grid, enabling crucial grid stability, peak shaving, and demand response functionalities.

Challenges in BMS design for microgrids encompass intricate system dynamics, dynamic load variations, intermittent renewable energy generation, and stringent grid synchronization requirements. Additionally, ensuring compatibility with diverse battery chemistries and addressing safety concerns in remote or off-grid installations are paramount considerations.

Advancements in BMS technology for microgrids include the development of advanced control algorithms for real-time energy management, predictive analytics for optimal dispatch strategies, and integration with smart grid communication protocols for enhanced grid interaction and control. Furthermore, innovative SoC estimation techniques and sophisticated cell balancing methodologies contribute to improved battery performance and longevity within microgrid applications.

This abstract underscores the vital role of BMS in facilitating the efficient integration and operation of battery energy storage systems within microgrids. It emphasizes ongoing research endeavors aimed at addressing the unique challenges encountered in microgrid environments and harnessing technological advancements to bolster the reliability, performance, and sustainability of microgrid energy systems

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1. Introduction

1.1. Microgrid

A microgrid represents a localized energy infrastructure utilizing diverse distributed resources for electricity generation and storage. In contrast to centralized grids, microgrids function autonomously, bolstering resilience and efficiency. They incorporate renewable sources such as solar panels and wind turbines, supported by sophisticated control mechanisms to maximize energy utilization. By fostering energy autonomy and sustainability, microgrids seamlessly integrate renewable energy into the larger framework of infrastructure. As demand surges for decentralized energy solutions, microgrids emerge as a compelling strategy to address contemporary energy dilemmas, promising enhanced resilience and efficiency while promoting renewable energy adoption.

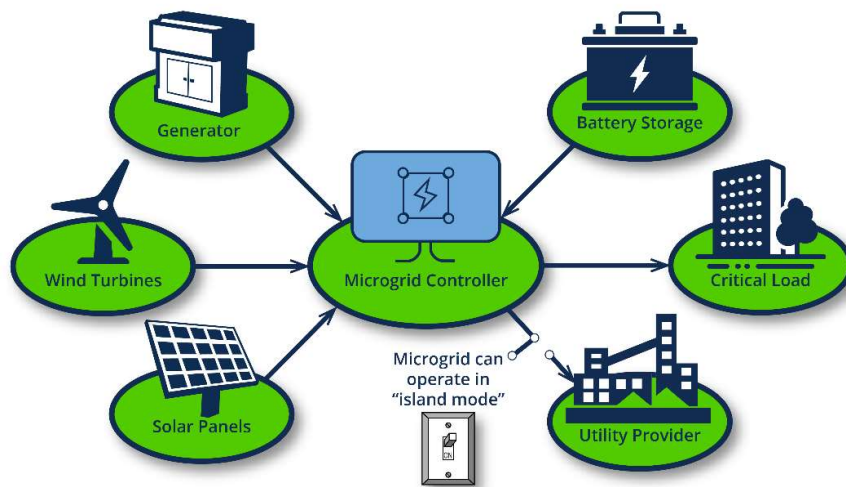


Figure 1.1: Microgrid Components

Components of Micro-grid

1. Renewable Energy Resources
2. Conventional Generators
3. Energy Storage Systems
4. Loads
5. Smart Systems and Sensors
6. Protection and Switching Equipment
7. Communication infrastructure

1.2. Battery Energy Storage System

Battery Energy Storage Systems (BESS) are highly versatile and increasingly popular components within microgrids due to their ability to provide various grid services and enhance overall system performance [1]. Here's how BESS contributes to microgrid functionality:

1. Energy Management and Peak Shaving.
2. Load Balancing and Frequency Regulation.
3. Voltage Support and Power Quality Improvement.
4. Grid Backup and Black Start Capability.
5. Renewable Energy Integration.
6. Grid Support Services and Ancillary Markets.
7. Remote and Islanded Operation.

Overall, BESS serves as a versatile and essential component within microgrids, offering a wide range of benefits including enhanced grid reliability, increased renewable energy penetration, improved energy efficiency, and participation in energy markets. As technology advances and costs continue to decline, BESS is expected to play an increasingly prominent role in the transition towards a more sustainable and resilient energy future [1].



Figure 1.2: Battery Energy Storage System

2. Battery Management System

A Battery Management System (BMS) is a crucial component of Battery Energy Storage Systems (BESS) and plays a vital role in ensuring the safe, efficient, and reliable operation of batteries within microgrids and other energy storage applications [2]. Here's an overview of its functions and importance:

1. State of Charge (SoC) Estimation.
2. State of Health (SoH) Monitoring.
3. Cell Balancing.
4. Temperature Management.
5. Voltage and Current Monitoring.
6. Safety and Fault Detection.
7. Control and Communication.
8. Energy Optimization.

In summary, a Battery Management System is essential for maximizing the performance, safety, and longevity of battery energy storage systems within microgrids and other applications. It provides comprehensive monitoring, control, and protection capabilities to ensure that batteries operate reliably and efficiently while meeting operational and safety requirements [2].

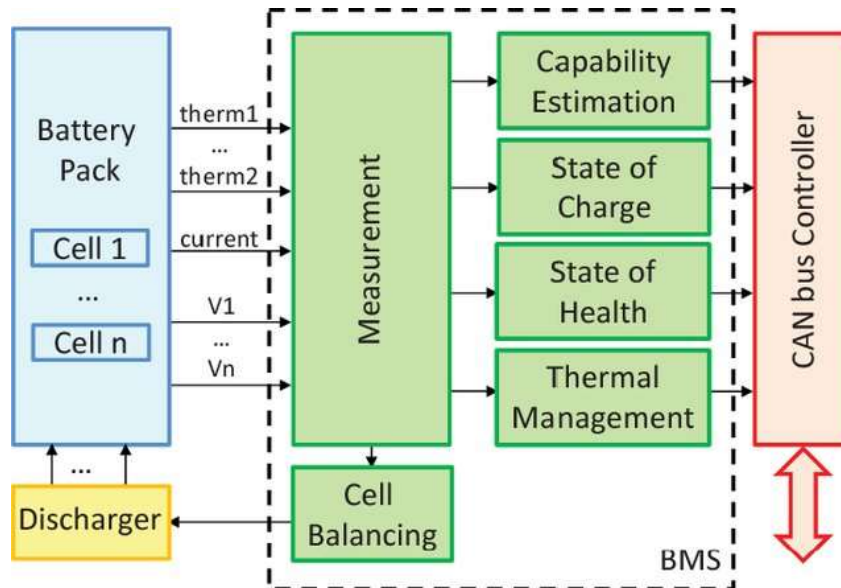


Figure 2.1: Block Diagram of BMS

2.1. Battery Specifications

Battery specifications and Conditions provide crucial information about the characteristics and performance of a battery. Here are key aspects covered in battery specifications and Conditions [3]:

Nominal Voltage:

The Nominal Voltage is the amount of voltage output cell gives out when charged.

Energy Density:

Energy Density is the measure of how much energy a battery contains in proportion to its weight.

Power Density:

Power Density of a battery is a measure of how fast the battery can deliver the energy.

Self-Discharge:

Self-Discharge is a phenomenon in batteries in which internal chemical reactions reduce the stored charge of the battery without any connection between the electrodes or any external circuit.

Life Cycle:

The Cycle Life of batteries is the number of charge and discharge cycles that a battery can complete before losing performance.

C - Rating:

It is a constant current charge or discharge rate, which the battery can sustain for one hour.

Depth Of Discharge:

Depth of Discharge (DOD) is the fraction or percentage of the capacity which has been removed from the fully charged battery.

Terminal Voltage:

The Voltage between the battery terminals with load applied. Terminal Voltage varies with SOC and Discharge/Charge current.

Understanding these specifications is essential for selecting the right battery for a particular application, ensuring optimal performance, longevity, and safety.

2.2. Classification of batteries

The data presented in the table highlights the advantageous characteristics of lithium-ion batteries in comparison to other types of batteries commonly utilized for energy storage purposes. Notably, lithium-ion batteries demonstrate superior energy density, energy efficiency, power density, and cycle life when contrasted with alternatives such as lead-acid, Ni-MH, and Ni-Cd batteries. These attributes collectively contribute to the widespread preference for lithium-ion batteries in energy storage applications.

While it is acknowledged that certain battery types, like Ni-MH batteries, may exhibit strengths in specific areas such as longer cycle life, the comprehensive performance profile of lithium-ion batteries remains unmatched. Additionally, the table underscores the potential of emerging battery technologies, such as sodium-sulphur (NaS), vanadium redox flow (VRB), and zinc-bromide batteries, which showcase promising characteristics such as high cycle life or low self-discharge rates. Nonetheless, the established track record and established infrastructure supporting lithium-ion technology continue to position it as the foremost choice for energy storage solutions.

Table 2.1: Comparison of different types of batteries

Type/Specifications	Energy Density (Wh/kg)	Energy Efficiency (%)	Power Density (W/kg)	Cycle Life (Cycles)	Self-Discharge (%/Month)
Lead-Acid	30-40	70-90	180	200-2000	3-4
Li-Ion	100-250	75-90	1800	500-2000	5-10
Li Polymer	130-200	70	3000	>1200	4-8
Ni-MH	30-80	70	250-1000	500-1000	30
Ni-Cd	40-60	60-90	140-180	500-2000	10-15
NaS	150	80-90	120-150	2500	-
VRB	25-40	80	100-150	>16000	<1
Zinc Bromide	70	70	-	1000	-

3. Methodology of Estimation of State of charge

Estimating the state of charge (SoC) of a battery is crucial for optimizing its performance, enhancing its lifespan, and ensuring safe operation. SoC refers to the current amount of charge remaining in a battery relative to its full capacity. By accurately estimating SoC, users can effectively manage energy consumption, plan charging cycles, and prevent over-discharge or overcharge, which can degrade battery health [5].

In various applications such as electric vehicles, portable electronics, and renewable energy systems, precise SoC estimation enables efficient utilization of battery resources, maximizing operational efficiency and extending device runtime. It also facilitates predictive maintenance, allowing users to anticipate battery replacement or servicing before critical failures occur.

Accurate SoC estimation relies on sophisticated algorithms that consider factors like voltage, current, temperature, and battery chemistry. These algorithms may incorporate machine learning techniques to improve accuracy over time by learning from historical data and usage patterns.

Overall, the purpose of estimating SoC is to optimize battery performance, prolong its lifespan, ensure safety, and enhance the overall efficiency and reliability of systems relying on battery power.

3.1. Open-Circuit Voltage Method

Open circuit voltage (OCV) estimation is a crucial aspect of comprehending a battery's state of charge, providing insights into its health and performance. OCV signifies the voltage present across a battery when it's disconnected and no current is flowing. This parameter is typically measured during periods of rest between charge and discharge cycles, allowing for a more accurate assessment of the battery's condition. Accurate OCV estimation is vital for optimizing battery performance, ensuring the reliability of energy storage systems, and facilitating effective management of available energy. By serving as a fundamental metric, OCV aids in gauging the available energy and overall health of a battery, enabling informed decision-making regarding its

usage and maintenance. OCV estimation doesn't require the battery to be actively charging or discharging.

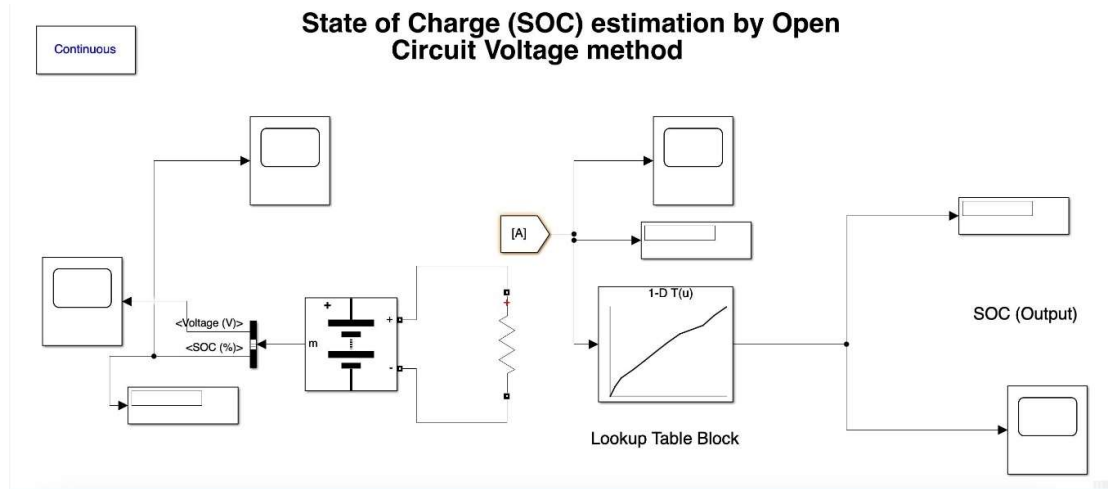


Figure 3.1: Simulink Model of Open Circuit Voltage method

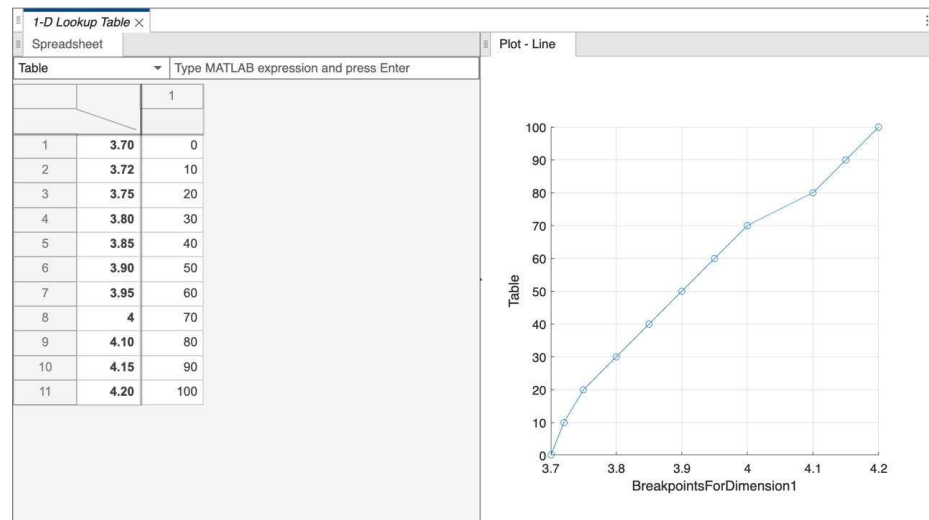


Figure 3.2: Look up table for Open Circuit voltage Method

In State of Charge (SOC) estimation methods, an Open Circuit Voltage (OCV) lookup table is used to correlate the battery's voltage with its SOC. This table contains voltage-SOC pairs obtained through calibration or modeling. By referencing this table, the SOC can be estimated based on the battery's voltage reading, allowing for a more accurate determination of the battery's state without the need for complex calculations or real-time modeling.

3.2. Coulomb Counting Method

Coulomb counting, a method utilized for assessing a battery's state of charge, integrates the current flow over time. This technique meticulously monitors the electrical charge entering and exiting the battery. By quantifying the cumulative charge, one can precisely gauge the remaining capacity and determine the state of charge. Widely adopted in battery management systems, Coulomb counting ensures meticulous monitoring and management of energy storage levels. This approach guarantees the optimization of battery performance, enhancing efficiency and reliability. By continuously tracking the flow of current, Coulomb counting enables proactive management of battery health, allowing for timely interventions and maintenance as needed. Overall, Coulomb counting serves as a fundamental tool in the realm of energy storage management, providing crucial insights into battery performance and longevity while facilitating efficient utilization of stored energy [6].

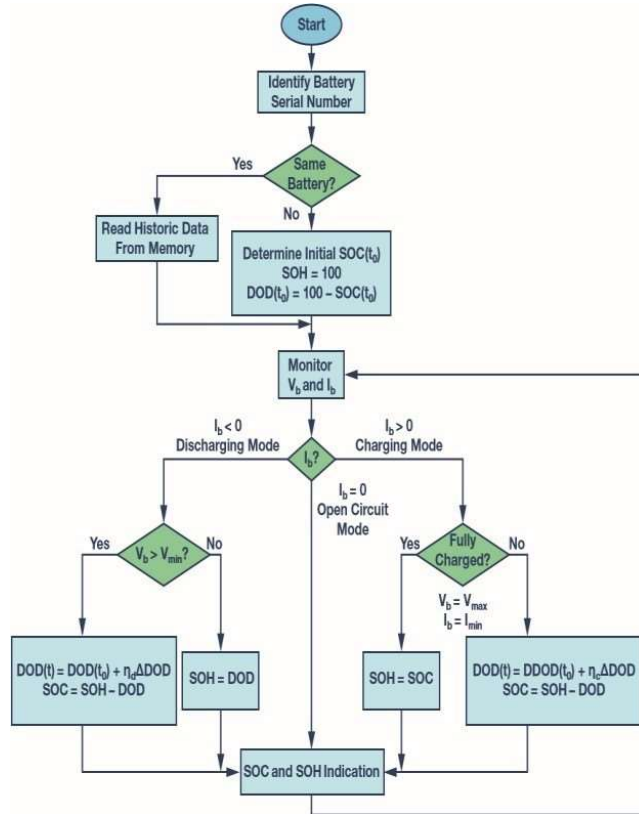


Figure 3.3: Flow Chart Diagram of Coulomb Counting Method [7].

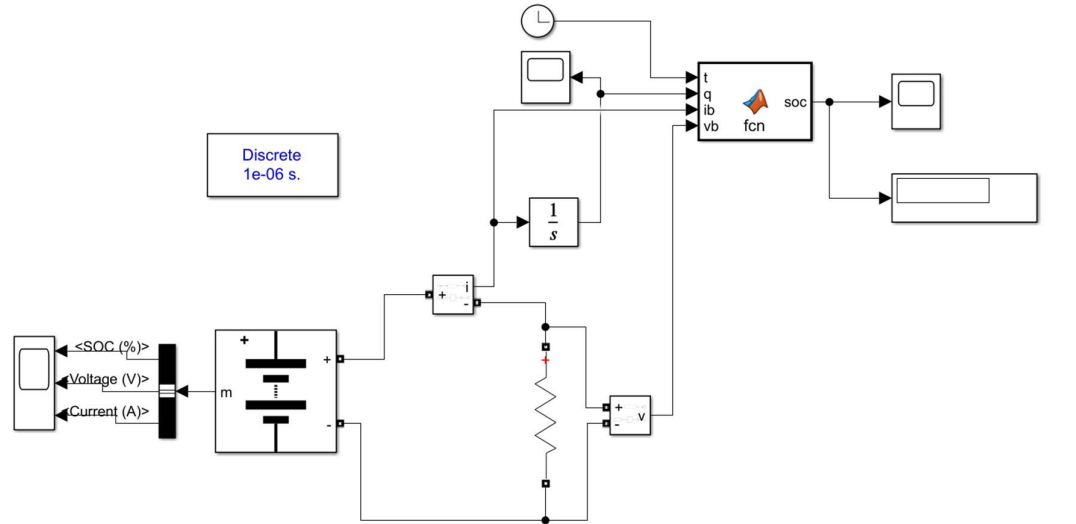


Figure 3.4: Simulink Model of Coulomb Counting Method

```
function soc = fcn(t, q, ib, vb)
i=0;
v=2.4;
if(ib==i)
    soc=100;
elseif(ib<0 || (ib>0&&vb>v))
    soc=80-(q/(2.08));
else
    soc=0;
end
```

Figure 3.5: Matlab Function Code for Coulomb Counting Method

3.3. Simulation Results

The dataset portrays the discharge characteristics of a lithium-ion battery, adhering to the standard nominal voltage of 3.2 volts for lithium-ion batteries. Throughout the discharge process, the initial state of charge (SoC) has been established at 80%. Implemented within a MATLAB function are specific directives that utilize both battery voltage and current, facilitating the display

of the State of Charge (SoC) during the battery discharge. There are losses during charging and discharging. These losses, in addition with the self-discharging, cause accumulating errors.

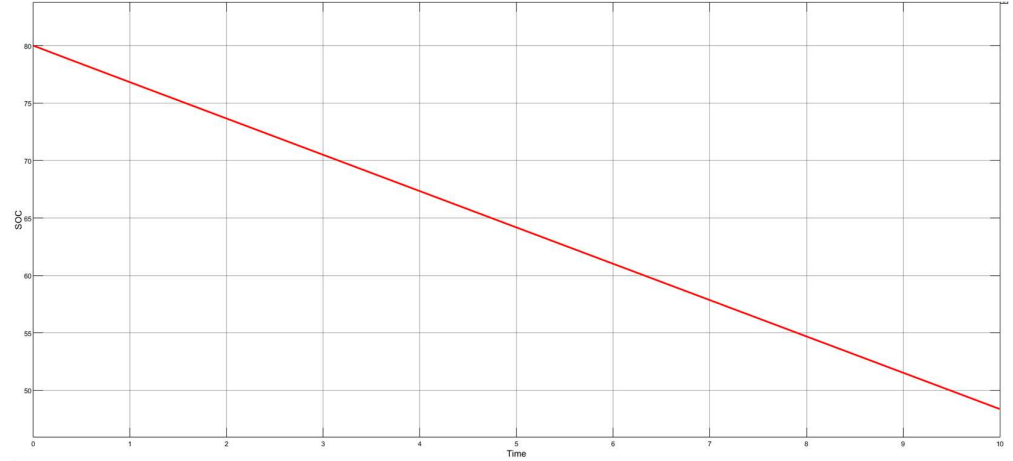


Figure 3.6: Simulation Result of Coulomb Counting Method

3.4. Kalman Filter Method

The Kalman filter is an estimation method widely used in various fields, including control systems and signal processing. It operates on a recursive algorithm that combines measurements and predictions to estimate the current state of a dynamic system. The Kalman filter excels in filtering out noise and providing accurate estimates, making it valuable in applications such as navigation, tracking, and sensor fusion. Its adaptability to changing conditions and ability to handle uncertain or incomplete information contribute to its effectiveness in real-time estimation tasks. Overall, the Kalman filter method is a powerful tool for enhancing the accuracy and reliability of state estimation in dynamic systems [8].

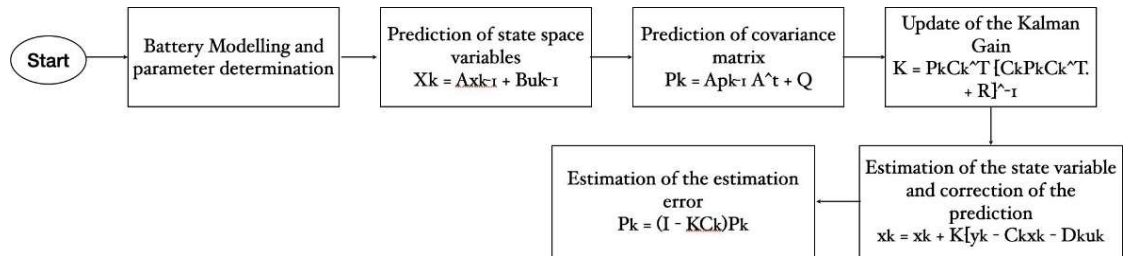
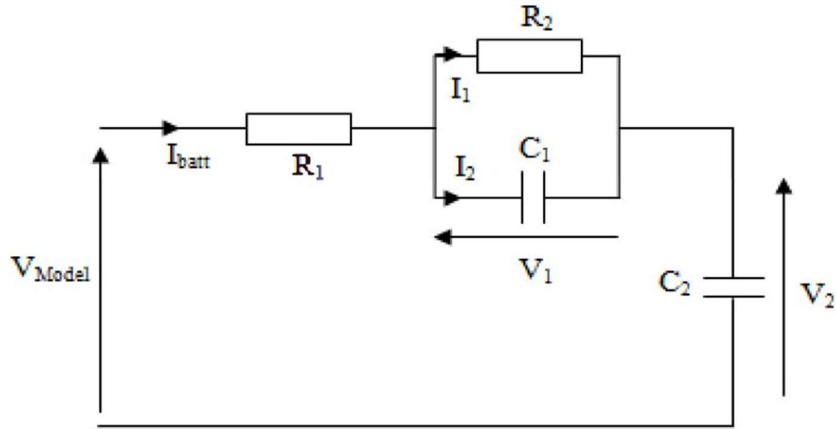


Figure 3.7: Flow Chart Diagram of Kalman Filter Method [8].

Li-ion Battery Modelling



Where,

R_1 : Internal resistance

C_1 : Double layer capacity

R_2 : Transfer resistance

C_2 : Battery capacity

V_1 : Voltage across the capacity

V_2 : Open circuit voltage

V_{model} : Model voltage

The following equation describes the relation between the open circuit voltage and SOC: [3]

$$V_2 = V_0 + g \cdot \text{SOC} \quad (1)$$

Where:

V_0 is the open circuit voltage when the battery is completely discharged, that is SOC= 0%.

$$\dot{V}_2 = I_{\text{batt}} / C_2 \quad (2)$$

The voltage V_1 across the capacitor C_1 is given by:

$$V_1 = R_2 * (I_{batt} - C_1 dV_1/dt) \quad (3)$$

$$V_{model} = V_2 + V_1 + R_1 * I_{batt} \quad (4)$$

The battery Model can be rewritten in a state space form as follows:

$$\dot{z} = Ax + Bu \quad (5)$$

$$y = Cx + Du$$

$$\text{with: } A = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{R_2 C_1} \end{bmatrix}, \quad B = \begin{bmatrix} -\frac{1}{g C_2} \\ \frac{1}{C_1} \end{bmatrix}, \quad C = [g \quad 1] \text{ and } D = R_1$$

The model state vector x consists of two state variables x_1 and x_2 , namely:

$$\begin{aligned} x_1 &= \{x_1 = \text{SOC} \\ x_2 &= V_1\} \end{aligned} \quad (6)$$

The system input u is the current I_{batt} and the output y is the voltage V_{model} :

$$\begin{bmatrix} \dot{\text{SOC}} \\ \dot{V}_1 \end{bmatrix} = \begin{bmatrix} \frac{I_{batt}}{g C b} \\ -\frac{V_1}{R_2 C b} + \frac{I_{batt}}{C_1} \end{bmatrix} \quad (7)$$

$$y = V_{model} = g * \text{SOC} + V_0 + V_1 + R_1 I_{batt} \quad (8)$$

These equations should be written in discrete time form to cope with real time requirements and implementation. To this end, we use a simple Euler discretization yielding.

$$\{X_{k+1} = A_d X_k + B_d u_k\} \quad (9)$$

$$Y_k = C_d X_k + D_d u_k\}$$

With:

$$A_d = \begin{bmatrix} 1 & 0 \\ 0 & 1 - \frac{\Delta t}{R_2 C_1} \end{bmatrix}, \quad B_d = \begin{bmatrix} -\frac{\Delta t}{g C_2} \\ \frac{\Delta t}{C_1} \end{bmatrix}, \quad C_d = C \text{ and } D_d = D$$

The system in discrete time form becomes:

$$\mathbf{x}_{k+1} = \begin{bmatrix} SOC_{k+1} \\ V1_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 - \frac{\Delta t}{R2C1} \end{bmatrix} \begin{bmatrix} SOC_k \\ V1_k \end{bmatrix} + \begin{bmatrix} -\frac{\Delta t}{gC} \\ \frac{\Delta t}{C1} \end{bmatrix} I_{batt} \quad (10)$$

Determination of the Model Parameters:

The parameters of the battery model have been extracted based on two discharge tests: constant current discharge (fig2) and pulse discharge (fig3). The value of the current in the constant discharge test is 0.1A during all the test time (10390s), (fig2) and the step discharge current (fig3) is only a step of current for 900s.

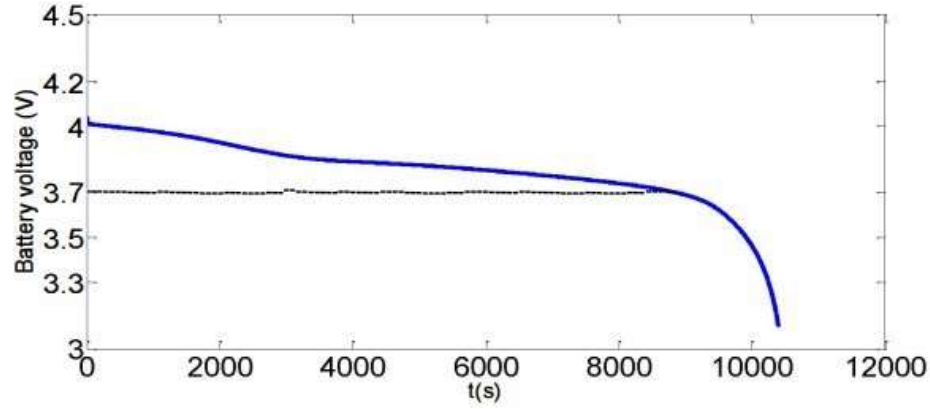


Figure 3.8: Constant Current Discharge Test

When the battery is completely discharged SOC=0%, the voltage is determined from the discharge curve of the battery.

The current starts to drop suddenly when the voltage reaches 3.7V. The battery voltage for SOC=0% is 3.7V and 4.2V for a fully charged cell

SOC=100%.

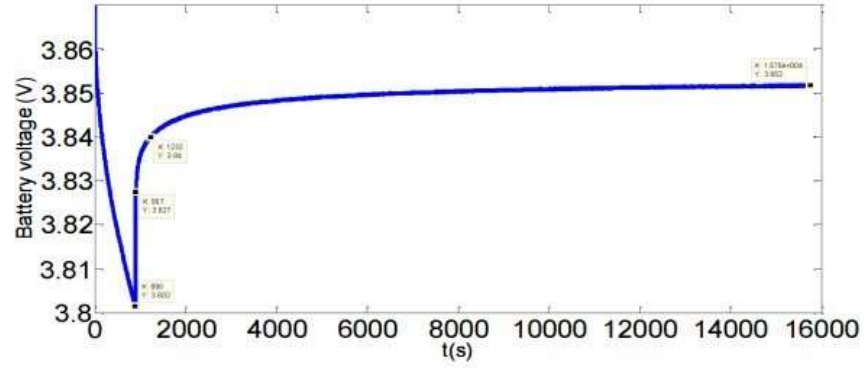


Figure 3.9: Pulse Discharge

The cell resistance R_1 is determined using the immediate rise of the voltage shown in Fig2, when the test circuit is open $I=0A$, divided by the value of the constant current discharge test $I=0.1 A$. Therefore, the resistance was determined as:

$$R_1 = \Delta V / I = (3.827 - 3.802) / 0.1 = 0.25 \Omega \quad (11)$$

The diffusion resistance R_2 of the cell is determined from the rise of battery voltage between the time of immediate increase and the time it reaches a pseudo-steady value during the rest period this voltage difference is divided by the constant current $I = 0.1A$ used for the discharge test.

Therefore, the diffusion resistance was determined as follows:

$$R_2 = \Delta V / I = (3.852 - 3.827) / 0.1 = 0.25 \Omega \quad (12)$$

The diffusing capacity C_1 is determined from the ratio between the time constant of the battery and the diffusion resistance.

From the curve, the time constant is determined as:

$$\tau = R_2 C_1 = 1232 - 907 = 325 \text{ s} \quad (13)$$

Therefore, diffusion capacity is:

$$C_1 = \tau / R_2 = 325 / 0.25 = 1300 \text{ F}$$

Table 3.1: Model Parameters

model parameters	R_1	R_2	C_1	C_2
Value	0.25Ω	0.25Ω	1300 F	8040 F

Important terms used in the code development of Kalman Filter:

SigmaW: (Process Noise Covariance)

This matrix describes how the different sources of noise in a system are correlated with each other and how much variance or uncertainty is associated with each noise source.

SigmaV: (Sensor Noise Covariance)

This matrix describes how the noise in sensor measurements is correlated between different sensors (if there are multiple sensors) and how much uncertainty or variance is associated with each measurement.

The SigmaV matrix is essential for filtering and estimation algorithms, such as the Kalman filter. It's used to account for the uncertainty in sensor measurements and to improve the accuracy of estimating the true state of a system in the presence of sensor noise.

SigmaX: (State Covariance Matrix)

This matrix is used to represent the initial covariance of the state estimation, indicating that there is no initial uncertainty or correlation between the state variables.

Code Flow:

Step 1: Initialization

$$\text{SigmaW} = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \text{ (Process noise covariance for Soc and Voltage)}$$

$$\text{SigmaV} = 0.1 \text{ (Sensor noise covariance for Soc and Voltage)}$$

State_true = [1; 4.2] (True initial state vector [Soc; Voltage])
State_hat = [1; 4.2] (Initial Kalman filter estimate for state vector)

$\text{SigmaX} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (Initial Kalman filter Covariance)

maxIter = 10 (Total Number of iterations)

u = 0 (Unknown initial driving input)

Step 2: Estimation of next state

State_hat =

$A * \text{State_hat} + B * u$

$\text{SigmaX} = A * \text{SigmaX} * A' + \text{SigmaW}$

$w = \text{chol}(\text{SigmaW}) * \text{rand}(2,1)$

$v = \text{chol}(\text{SigmaV}) * \text{rand}(1)$

Step 3: Calculation of true system output

$z_{\text{true}} = C * \text{State_true} + D * u + v$

$\text{State_true} = A * \text{State_true} + B * u + w$ (next true state vector)

Step 4: Estimation of System output

$z_{\text{hat}} = C * \text{State_hat} + D$

Step 5: Computation of kalman gain

$\text{SigmaZ} =$

$C * \text{SigmaX} * C' + \text{SigmaV}$

$L = \text{SigmaX} * C' / \text{SigmaZ}$

Step 6: Updating of the estimated value with Kalman gain

$$\text{State_hat} = \text{State_hat} + L * (z_{\text{true}} - z_{\text{hat}})$$

Step 7: Error covariance measurement update

$$\text{SigmaX} = \text{SigmaX} - L * \text{SigmaZ} * L'$$

Step 8: Calculation of the state estimation error:

$$\text{State_error} = \text{State_hat} - \text{State_true}$$

Step 3 to Step 8 will repeat till total number of iteration (10 in this case)

3.5. Simulation Results

The provided plots depict the convergence of a Kalman filter over 10 iterations in estimating state of charge (SoC) and voltage values. The first plot shows iterative refinement of SoC estimates, approaching true values by minimizing disparities with measurements. The second plot illustrates dynamic adjustment of voltage estimates, converging towards true values with assimilated measurements. The third plot displays diminishing voltage estimation errors, highlighting the filter's ability to optimize estimates by balancing system dynamics and measurement errors. These plots collectively demonstrate the Kalman filter's iterative and adaptive nature, progressively enhancing precision and reliability in state and voltage estimations over time.

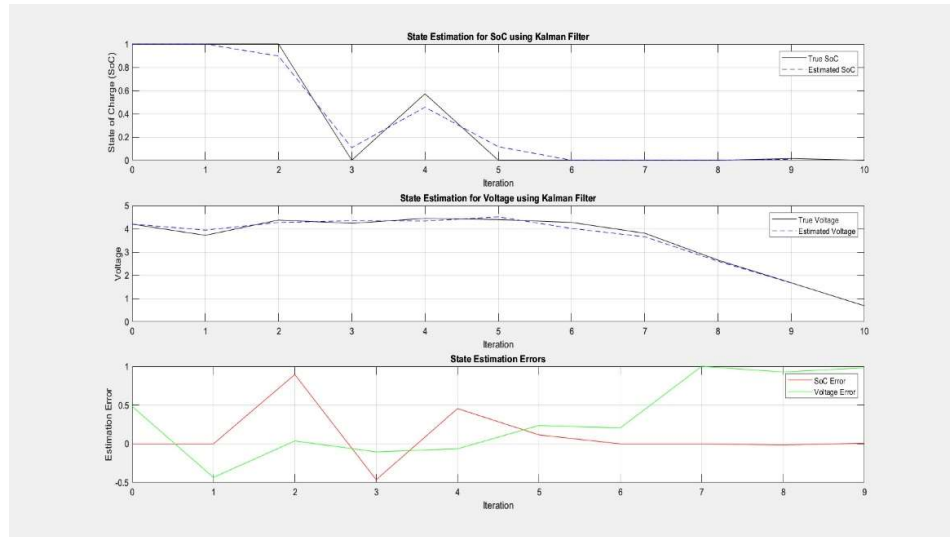


Figure 3.10: SoC Estimation using Kalman filter method.

4. Methodology of Cell Balancing

Cell balancing is a critical aspect of managing multi-cell battery packs to ensure that individual cells within the pack are maintained at similar voltage levels and states of charge (SoC). Cell balancing helps maximize the energy storage capacity, extend battery life, and improve overall system performance.

Effective cell balancing is essential for maximizing the performance, safety, and lifespan of battery packs in various applications, including electric vehicles, renewable energy systems, and microgrids. The choice of cell balancing technique depends on factors such as pack size, cell chemistry, cost constraints, and performance requirements.

4.1. Passive Cell Balancing

Passive cell balancing methods are essential techniques employed in managing the state of charge among individual cells within a battery pack. Unlike active methods that involve transferring energy between cells, passive balancing relies on dissipative elements, such as resistors, to redistribute excess charge. This process occurs naturally during charging and discharging, ensuring that cells remain within a specified voltage range. Passive balancing is known for its simplicity, cost-effectiveness, and reliability, as it doesn't require additional control circuitry. While it may be less efficient than active methods, passive cell balancing is widely used in various energy storage systems to enhance overall battery performance and lifespan [9].

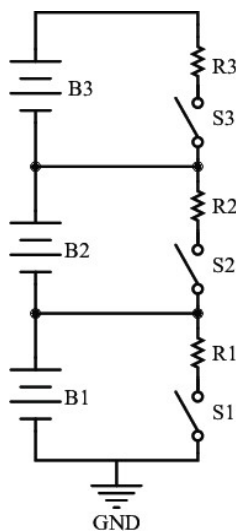


Figure 4.1: Circuit Diagram of Passive Cell Balancing

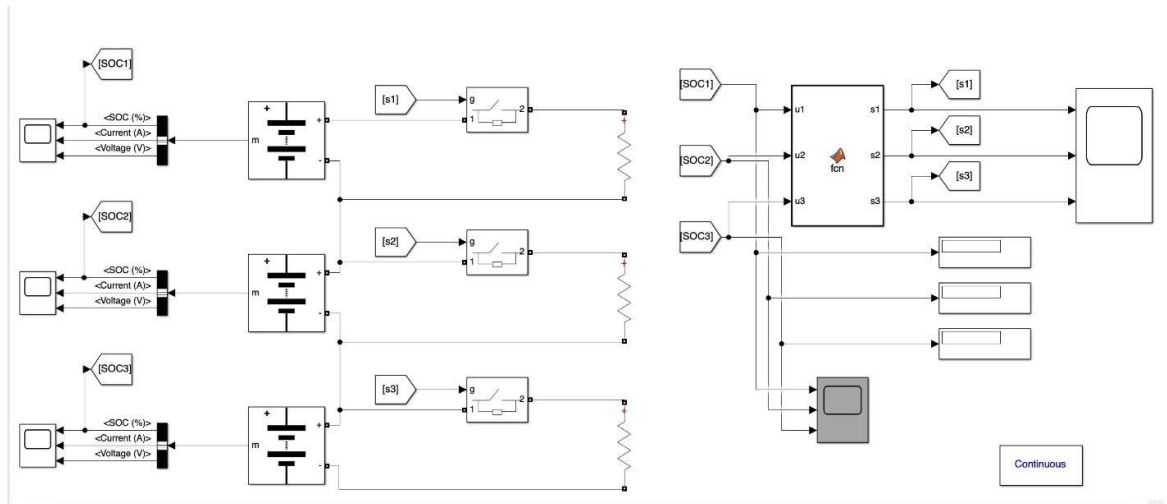


Figure 4.2: Simulink Model of Passive Cell Balancing Circuit

```
function [s1,s2,s3] = fcn(u1,u2,u3)

u1 = int32(u1);
u2 = int32(u2);
u3 = int32(u3);

if((u1>u2) || (u1>u3))
    s1=1;
else
    s1=0;
end
if((u2>u1) || (u2>u3))
    s2=1;
else
    s2=0;
end
if((u3>u1) || (u3>u2))
    s3=1;
else
    s3=0;
end
```

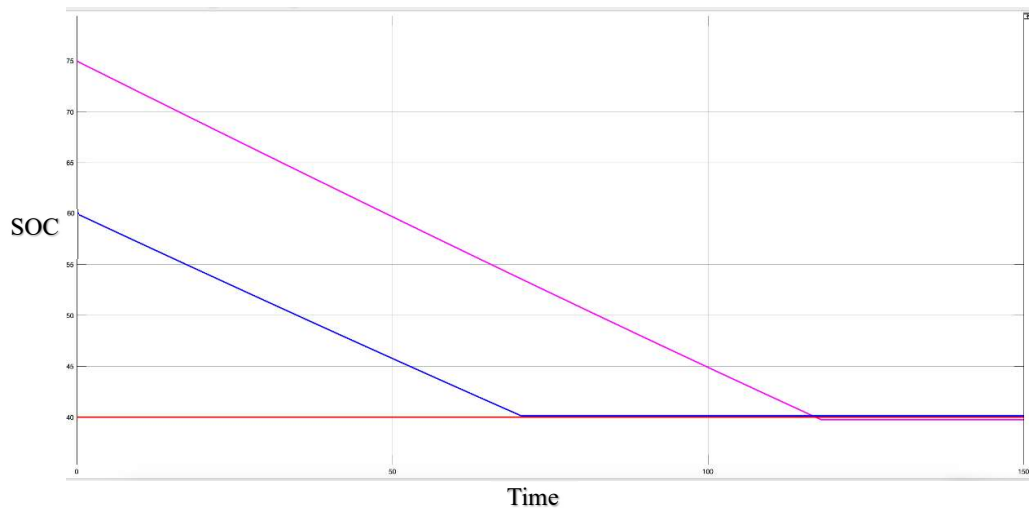
Figure 4.3: Matlab function Code for Passive Cell Balancing Circuit

Table 4.1: Parameters of Passive Cell Balancing Circuit

Parameters	Value
Nominal Voltage of 3 Cells (V)	3.7 V
Rated Capacity (Ah)	5.4 Ah
SoC of Cell 1 (%)	75%
SoC of Cell 2 (%)	40%
SoC of Cell 3 (%)	60%
Resistance (Ω)	1 Ω

4.2. Simulation Results

Based on the Simulink model of Passive balancing, a MATLAB code is generated using MATLAB function where the SoCs of the entire cells are compared and its output is fed to the gate of the switch thereby controls the operation of the circuit. Three cells are given different SoCs. Cell 1 SoC is 75%, Cell 2 is 40%, Cell 3 is 60%. Out of the three cells, cell 2 is having minimum SoC level. First second and third cell discharges through a Resistor of 1 Ohm whose Nominal voltage is 3.6V, rated capacity 5.4Ah and it balances to around 40% which is the SoC of the second cell. Fig-4.2 shows the simulation result of Passive balancing. Out of the three batteries second battery having lower SoC of 40%, So first battery whose SoC1 discharges from 75% to 40%, Third battery whose SoC is 60% discharges and reach 40%. Second battery continues to be in around 40%, i.e., as time progresses, we observe that all cells are discharged to match the cell with the lowest SoC.

**Figure 4.4:** SoC Levels of all three Cells after Cell Balancing.

4.3. Active Cell Balancing

Active cell balancing in Battery Management Systems (BMS) actively equalizes the voltage and state of charge (SoC) of individual cells, employing techniques that transfer energy between cells using circuits like DC-DC or flyback converters. Unlike passive methods, which dissipate excess energy as heat, this approach maximizes battery pack capacity and lifespan. It prevents issues such as overcharging or deep discharging, particularly important for applications with multiple cells like electric vehicles and renewable energy systems. Achieving uniform voltage and SoC across cells optimizes overall performance, efficiency, and reliability, thereby enhancing system longevity and functionality.

Active cell balancing is particularly beneficial in applications where battery packs consist of multiple cells connected in series or parallel, such as electric vehicles, renewable energy systems, and grid-scale energy storage. By ensuring uniform voltage and SoC across all cells, active cell balancing helps optimize the overall performance, efficiency, and reliability of the battery pack, ultimately improving the performance and longevity of the entire system [9].

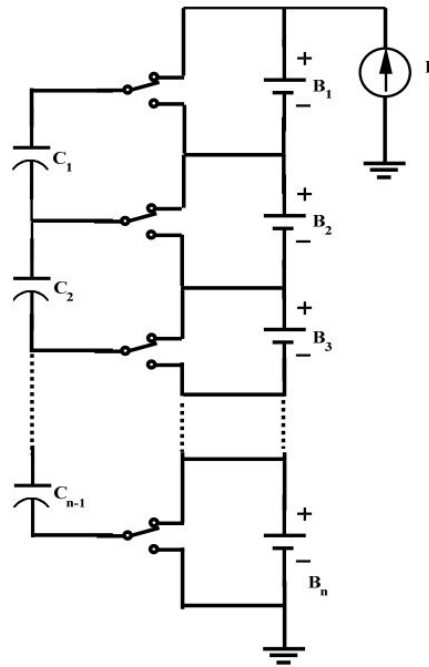


Figure 4.5: Circuit Diagram of Active Cell Balancing

4.3.1. Switched Inductor Cell Balancing

Switched inductor cell balancing is a technique used in Battery Management Systems (BMS) to actively balance the voltage of individual cells within a battery pack. It involves the use of switched inductor circuits to transfer energy between cells, effectively equalizing their voltage levels.

Working of Switched Inductor Cell Balancing:

Switched inductor cell balancing is a technique used in Battery Management Systems (BMS) to actively balance the voltage of individual cells within a battery pack. It involves the use of switched inductor circuits to transfer energy between cells, effectively equalizing their voltage levels.

Here's how switched inductor cell balancing typically works:

1. **Monitoring:** The BMS continuously monitors the voltage of each cell within the battery pack to identify any imbalances. Cells with higher voltage levels are targeted for balancing.
2. **Switched Inductor Circuits:** Each cell is connected to a switched inductor circuit, which consists of an inductor and a switching element (such as a transistor or MOSFET).
3. **Energy Transfer:** When a cell requires balancing (i.e., its voltage is higher than desired), the switching element in the corresponding switched inductor circuit is activated. This allows current to flow through the inductor, storing energy in its magnetic field.
4. **Energy Redistribution:** Once the inductor has stored sufficient energy, the switching element is deactivated, causing the stored energy to be released back into the battery pack. This energy is transferred to cells with lower voltage levels, effectively raising their voltage and equalizing the overall pack voltage.

5. Control Algorithm: The BMS employs a control algorithm to determine when and how much energy should be transferred between cells to achieve voltage balancing. The algorithm takes into account factors such as cell voltages, pack configuration, and balancing priorities.

Switched inductor cell balancing offers several advantages, including high efficiency, fast response times, and reduced heat generation compared to passive balancing techniques. However, it may require additional circuitry and control complexity, which can increase system cost and complexity.

Overall, switched inductor cell balancing is an effective method for maintaining voltage balance within battery packs, thereby improving overall performance, efficiency, and lifespan of the battery system.

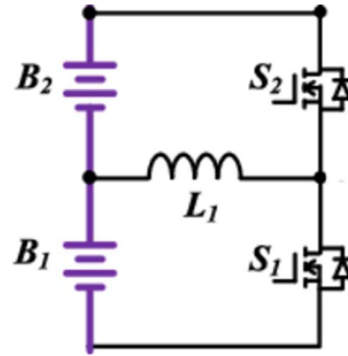


Figure 4.6: Circuit Diagram of Switched Inductor Active Cell Balancing

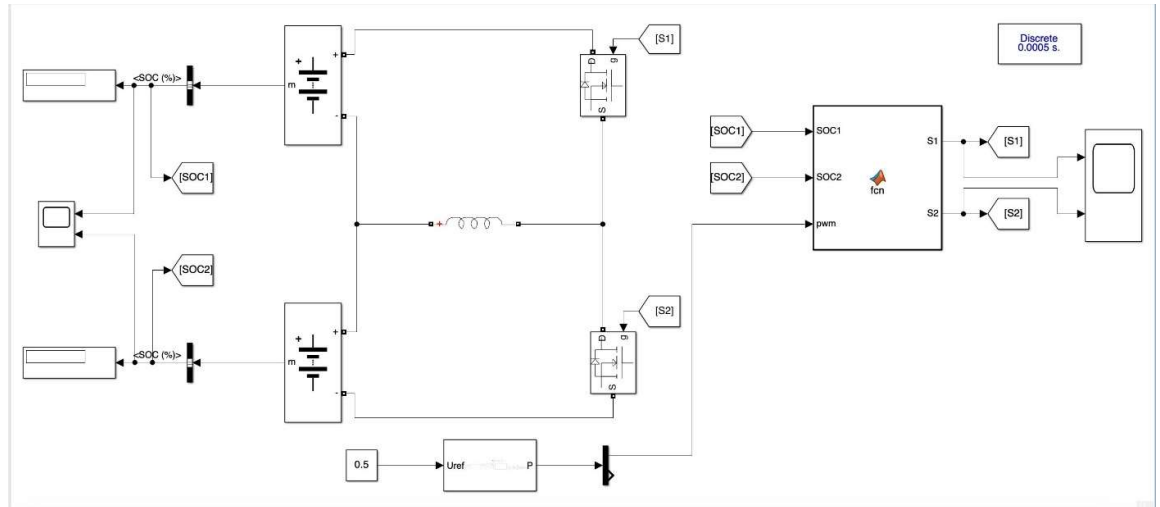


Figure 4.7: Simulink Model of Switched Inductor Active Cell Balancing Circuit

```

function [S1,S2] = fcn(SOC1,SOC2,pwm)

if (SOC2<SOC1)
    S1=pwm;
    S2=0;

elseif (SOC1<SOC2)
    S1=0;
    S2=pwm;

else
    S1=0;
    S2=0;

end

```

Figure 4.8: Matlab Function Code for Switched Inductor Active Cell Balancing Circuit

Table 4.2: Parameters of Switched Inductor Active Cell Balancing Circuit

Parameters	Value
Nominal Voltage of 2 Cells (V)	3.7 V
Rated Capacity (Ah)	5.4 Ah
SoC of Cell 1 (%)	75%
SoC of Cell 2 (%)	40%
Inductance (H)	1mH

4.3.2. Simulation Results

Utilizing a Simulink model, a MATLAB code has been generated to implement Switched Inductor Cell Balancing. The MATLAB function compares the State of Charges (SoCs) of all cells in the battery pack. The output of this comparison is then utilized to control the operation of the circuit by feeding it to the gate of the switch.

In this scenario, two cells are initialized with different SoC levels: Cell 1 with a SoC of 75%, and Cell 2 with a SoC of 40%. Notably, Cell 2 exhibits the minimum SoC level. The operational mechanism involves the second cell storing charge in an inductor. The inductor has a nominal voltage of 3.6V and a rated capacity of 5.4Ah. Through this process, the SoC of Cell 2 is balanced to approximately 48.5%.

The simulation results, as depicted in Fig-4.5, illustrate the efficacy of the Switched Inductor Active balancing technique in equalizing SoC levels among the battery cells. This approach facilitates efficient charge redistribution, ensuring uniformity across the battery pack and enhancing overall performance and longevity.

Additionally, the use of MATLAB code enables precise control and optimization of the cell balancing process, contributing to improved energy utilization and system reliability. Furthermore, the utilization of switched inductors offers advantages such as low power losses and reduced complexity compared to other balancing methods.

In summary, the integration of Simulink modeling, MATLAB code generation, and Switched Inductor Cell Balancing represents a robust approach for achieving optimal battery pack performance in various applications, ranging from electric vehicles to renewable energy storage systems.

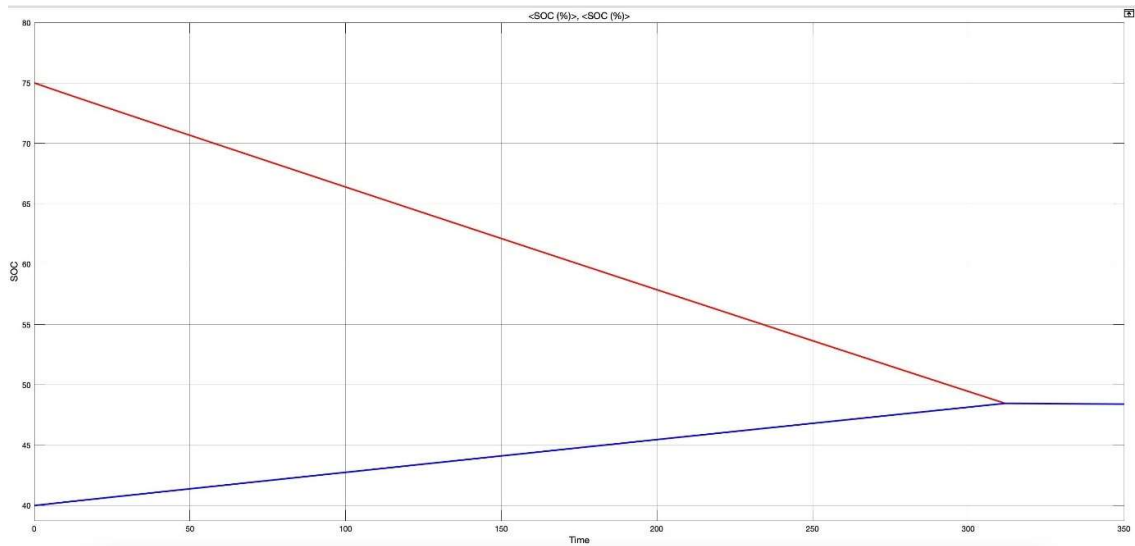


Figure 4.9: SoC Levels of all two Cells after Cell Balancing.

4.3.3. Switched Capacitor Cell Balancing

Switched capacitor cell balancing is a technique used in Battery Management Systems (BMS) to actively balance the voltage of individual cells within a battery pack. It involves the use of switched capacitor circuits to transfer charge between cells, effectively equalizing their voltage levels [9].

Working of Switched Capacitor Cell Balancing:

1. **Monitoring:** The BMS continuously monitors the voltage of each cell within the battery pack to identify any imbalances. Cells with higher voltage levels are targeted for balancing.
2. **Switched Capacitor Circuits:** Each cell is connected to a switched capacitor circuit, which consists of capacitors and switches. These circuits are used to transfer charge between cells.
3. **Charge Transfer:** When a cell requires balancing (i.e., its voltage is higher than desired), the switched capacitor circuit is activated. This causes charge to be transferred from the higher voltage cell to the lower voltage cell through the capacitors.
4. **Voltage Equalization:** As charge is transferred between cells, their voltages start to equalize. The process continues until the voltages of all cells reach a desired level, effectively balancing the battery pack.
5. **Control Algorithm:** The BMS employs a control algorithm to determine when and how much charge should be transferred between cells to achieve voltage balancing. The algorithm takes into account factors such as cell voltages, pack configuration, and balancing priorities.

Switched capacitor cell balancing offers several advantages, including high efficiency, fast response times, and reduced heat generation compared to

passive balancing techniques. However, it may require additional circuitry and control complexity, which can increase system cost and complexity.

Overall, switched capacitor cell balancing is an effective method for maintaining voltage balance within battery packs, thereby improving overall performance, efficiency, and lifespan of the battery system.

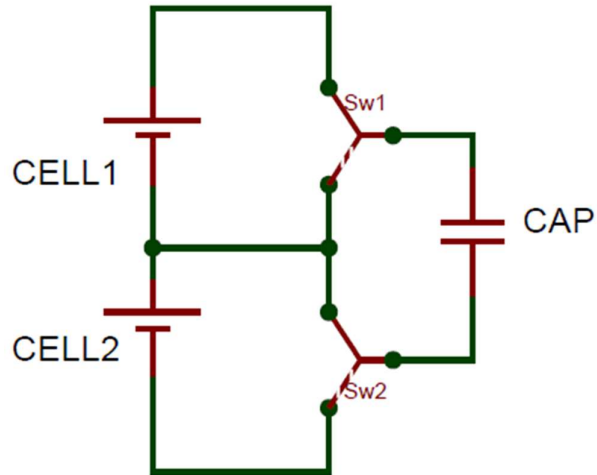


Figure 4.10: Circuit Diagram of Switched Capacitor Active Cell Balancing

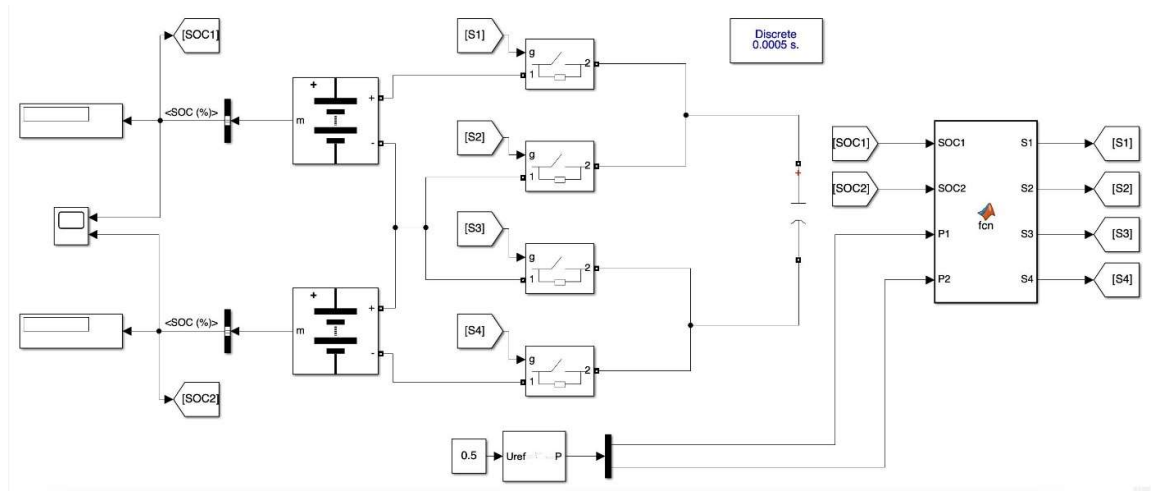


Figure 4.11: Simulink Model of Switched Capacitor Active Cell Balancing Circuit

```

cellbalancing2cells ▶ MATLAB Function

function [S1,S2,S3,S4] = fcn(SOC1,SOC2,P1,P2)

if (SOC2<SOC1)
    S1=P1;
    S2=P2;
    S3=P1;
    S4=P2;
elseif (SOC1<SOC2)
    S1=P1;
    S2=P2;
    S3=P1;
    S4=P2;
else
    S1=0;
    S2=0;
    S3=0;
    S4=0;
end

```

Figure 4.12: Matlab Function code for Switched Capacitor Active Cell Balancing Circuit

Table 4.3: Parameters of Switched Capacitor Active Cell Balancing Circuit

Parameters	Value
Nominal Voltage of 3 Cells (V)	3.7 V
Rated Capacity (Ah)	5.4 Ah
SoC of Cell 1	75%
SoC of Cell 2	40%
Capacitance (F)	0.1 F

4.3.4. Simulation Results

Upon examining the simulation outcomes of the switched capacitor cell balancing, a notable observation emerges regarding the initial State of Charge (SoC) levels of the two cells. Initially, Cell 1 exhibits a SoC of 75%, whereas the other cell registers a SoC of 40%. Following the cell balancing procedure, it becomes evident that both cells converge to a SoC level of 57.5%, representing the arithmetic average of the initial SoC levels with no discernible losses.

This observation underscores the effectiveness of the switched capacitor cell balancing technique in equalizing the SoC levels of disparate cells within the battery pack. By redistributing charge between cells, the balancing process successfully mitigates the initial SoC imbalance, ensuring uniformity across the battery pack.

Furthermore, the ability to achieve a balanced SoC state without incurring losses highlights the efficiency of the switched capacitor approach. This efficiency is crucial for maximizing the energy utilization of the battery pack and prolonging its operational lifespan.

In summary, the simulation results reaffirm the efficacy of switched capacitor cell balancing in harmonizing SoC levels among battery cells, thus enhancing overall battery pack performance and longevity.

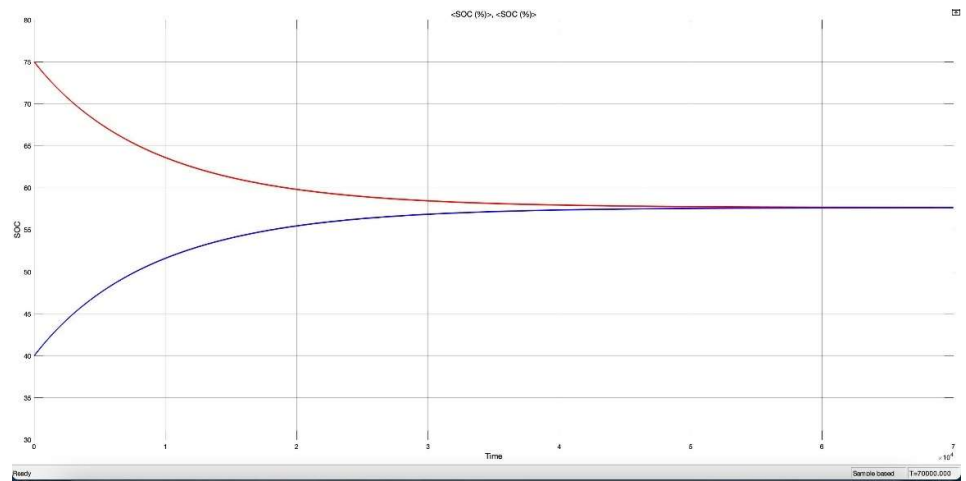


Figure 4.13: SoC Levels of all two Cells after Cell Balancing.

4.3.5. Comparative Analysis of Switched Inductor and Switched Capacitor Cell Balancing

Based on the simulated outcomes of both switched inductor and switched capacitor cell balancing techniques, as depicted from the figure-4.5 and figure 4.7, a discernible trend emerges: the State of Charge (SoC) levels of individual cells are consistently higher in capacitor cell balancing compared to inductor cell balancing. This observation holds true throughout the duration of the simulation, indicating a robust advantage of capacitor-based balancing methods in maintaining elevated SoC levels.

Post-balancing, the SoC levels of both cells in the capacitor balancing scenario stabilize around 57.5%, representing a substantial increase from their initial levels. Conversely, in the inductor balancing scenario, the cells' SoC levels settle at approximately 48.5% post-balancing, reflecting a comparatively lower improvement.

Furthermore, the capacitor cell balancing technique demonstrates a more rapid and efficient redistribution of charge among cells, leading to quicker convergence towards desired SoC levels. This efficiency is attributed to the inherent characteristics of capacitors, which offer low resistance pathways for charge transfer.

Moreover, the capacitor-based approach exhibits greater scalability and ease of implementation compared to its inductor-based counterpart. The simplicity of capacitor-based circuitry allows for streamlined integration into battery management systems, reducing complexity and potential points of failure.

In conclusion, the comprehensive analysis of simulation results supports the conclusion that capacitor cell balancing surpasses inductor cell balancing in terms of effectiveness, efficiency, and practicality. This advantage

underscores the potential of capacitor-based techniques as preferred solutions for optimizing battery pack performance and longevity in various applications.

4.3.6. Comparative analysis of Passive Cell Balancing and Active Cell Balancing

Analysis of simulation outcomes comparing passive and active cell balancing techniques reveals notable differences in State of Charge (SoC) levels between the two methodologies. In the active cell balancing scenario, both cells exhibit significantly higher SoC levels post-balancing, with levels stabilizing around 57.5%. Conversely, cells subjected to passive cell balancing reach markedly lower SoC levels post-balancing, hovering around 40% [9].

This discrepancy underscores the superior efficacy of active cell balancing over passive methods in elevating SoC levels within battery cells. Active cell balancing systems dynamically monitor and adjust individual cell voltages, facilitating optimal charge redistribution and maintaining balanced SoC levels across the battery pack. This proactive approach enables more efficient redistribution of charge compared to passive techniques.

Passive cell balancing, while simpler and potentially more cost-effective, relies on resistive-based methods for charge equalization. However, these methods may struggle to achieve the same level of charge redistribution efficiency as active counterparts due to their reliance on passive components.

The observed disparity in SoC levels highlights the importance of dynamic intervention and active management strategies in optimizing battery pack performance and longevity. Consequently, the findings suggest that active cell balancing, particularly when utilizing capacitor-based techniques, offers superior efficiency over passive cell balancing methods.

In conclusion, the simulation results support the assertion that capacitor cell balancing exhibits greater efficacy than inductor cell balancing, particularly within the context of active cell balancing strategies. This underscores the significance of employing dynamic and interventionist approaches to achieve optimal charge redistribution and ensure balanced SoC levels across battery cells [9].

4.3.7. Inductor based Cell Balancing technique based on Buck-Boost Topology

This method introduces a novel current-mode controller for switched-inductor topology to achieve voltage balancing among battery cells or modules within a pack while enhancing operational efficiency. It addresses drawbacks of fixed duty cycle schemes by offering precise control of balancing current, mitigating concerns about component current ratings and system parasitic elements. The selection of balancing current reference is optimized for maximum efficiency, minimizing power losses.

Furthermore, the method outlines a cell balancing algorithm tailored for implementation alongside the controller, focusing on achieving open circuit voltage balancing for enhanced accuracy in state-of-charge assessment compared to terminal voltage measurements.

Experimental results from a laboratory prototype featuring a three-cell battery pack substantiate the efficacy of the proposed current control and balancing method. These results comprehensively evaluate performance across various modes, providing empirical validation of effectiveness and efficiency in real-world applications.

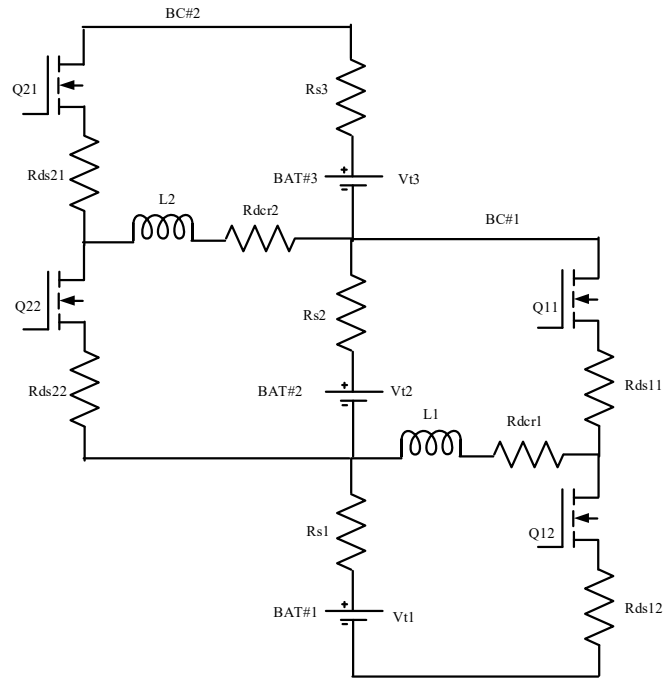


Figure 4.14: Circuit Diagram of Inductor-based cell Balancing based on buck-boost topology [10].

Table 4.4: Balancing Converter Design Parameters [10].

Converter Parameter	Value
Inductance L	33 μ H
Inductor DC Resistance R _{dc}	2.6m Ω
MOSFET ON Resistance R _{ds}	16m Ω
Switching Frequency f _{sw}	100kHz
Nominal Voltage of 3 cells	3.7V

Operation principle of the balancing controller:

The illustration of the current controller designed for battery voltage balancing is displayed in Fig. 4.9, intended for a battery pack consisting of n battery cells. As depicted in the figure, there are n-1 current control loops corresponding to the n battery cells. Each inductor current is precisely regulated to optimize the efficiency of the Battery Converter (BC). The direction of each inductor current is contingent upon whether the upper or lower cell of each BC, referring to each pair of adjacent battery cells, possesses a higher voltage. This determination is indicated by the local voltage difference $\Delta V_{m+1, m}$ for every two adjacent cells within each BC. In the case of a three-cell system, the local voltage differences are expressed as $\Delta V_{21} = V_{t2} - V_{t1}$ (the voltage difference between BAT#2 and BAT#1) and $\Delta V_{32} = V_{t3} - V_{t2}$ (the voltage difference between BAT#3 and BAT#2). [10].

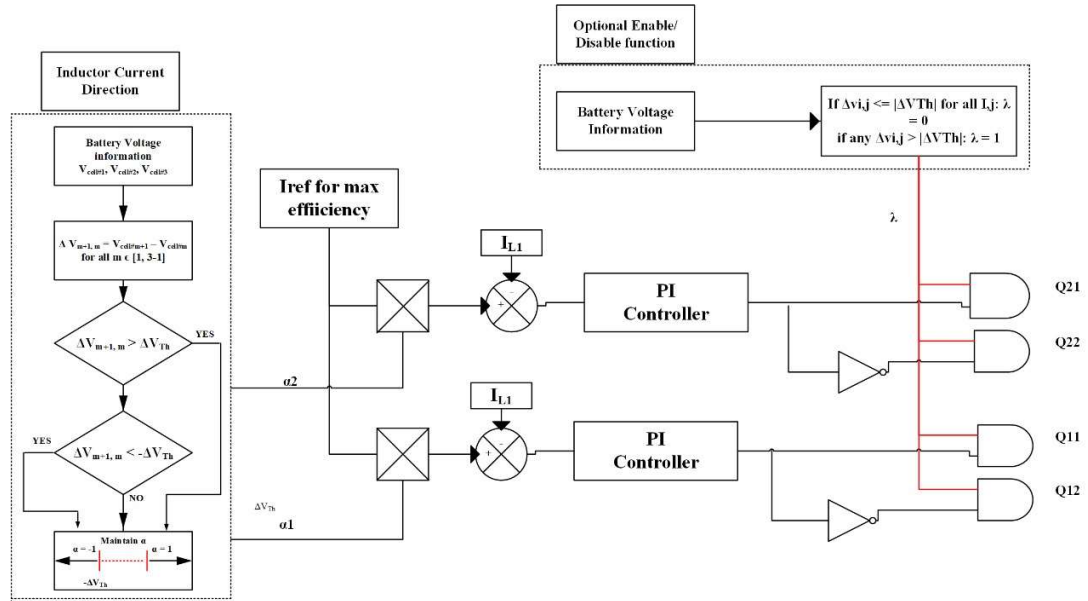


Figure 4.15: Diagram of the presented controller for voltage balancing [10].

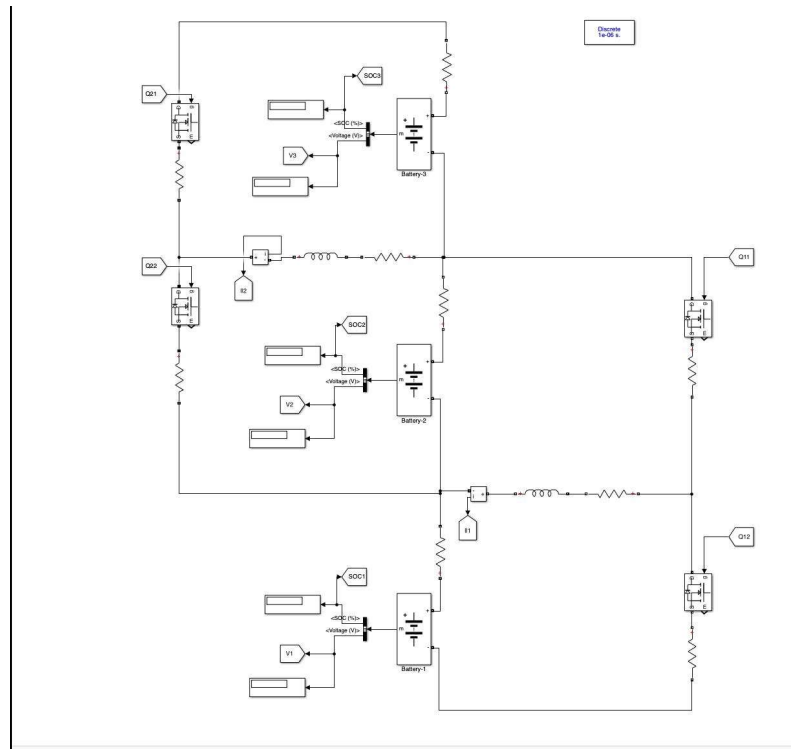


Figure 4.16: Simulink Model of the presented Controller.

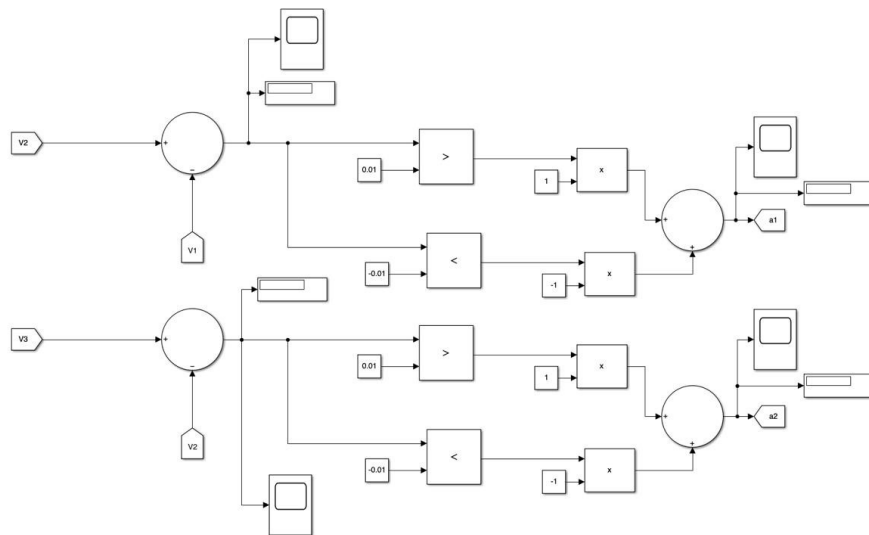


Figure 4.17: Simulink Model of the Algorithm for value α (α).

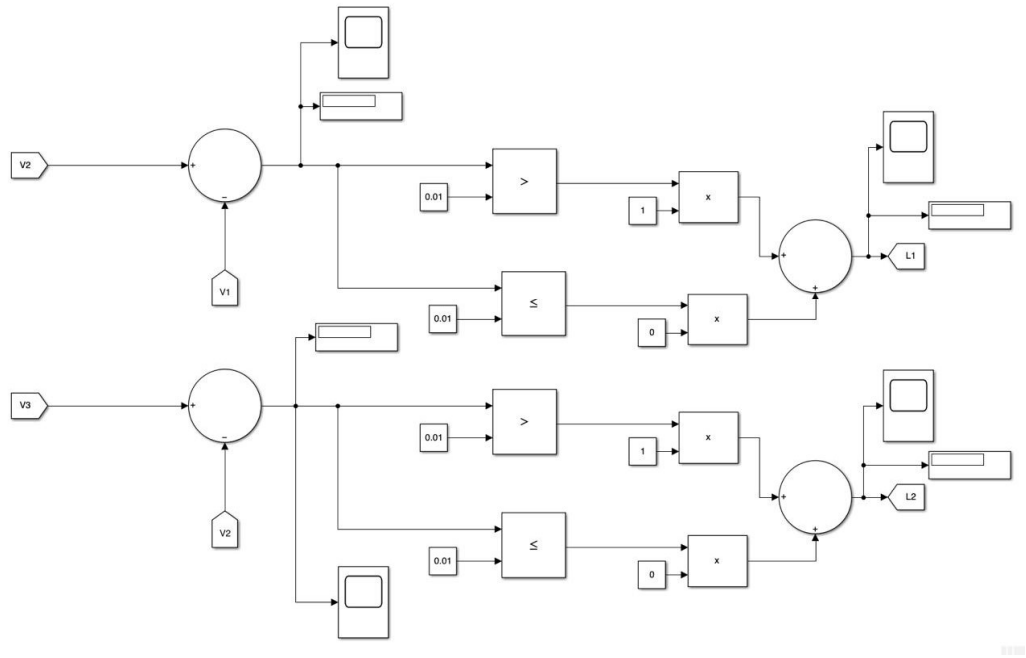


Figure 4.18: Simulink Model of the Algorithm for value Lambda (λ).

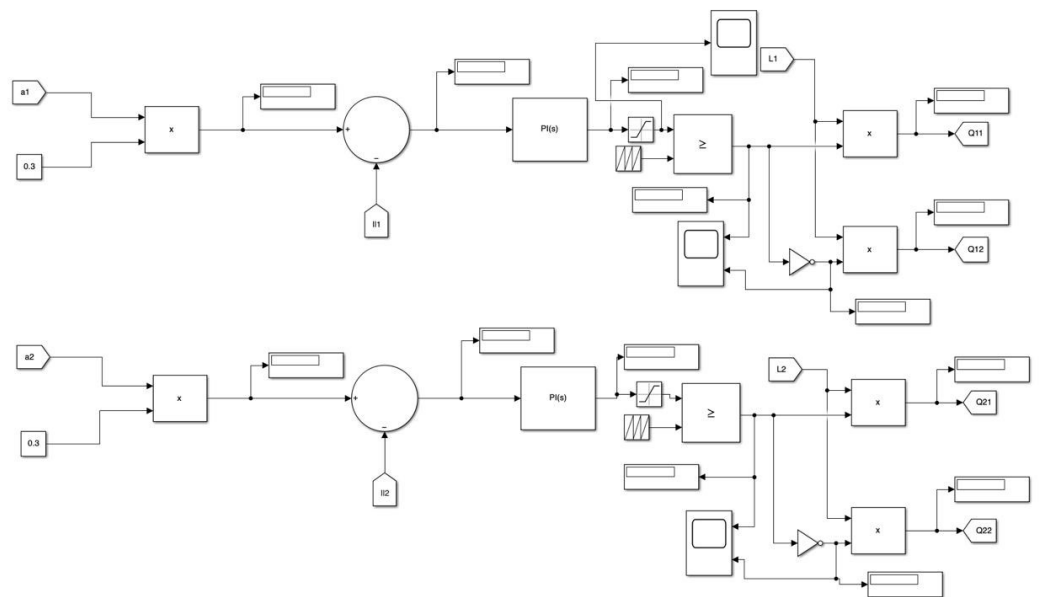


Figure 4.19: Simulink Model of the Algorithm for Switching Operation.

The algorithm governing the direction of each inductor current is depicted in the left section of Fig. 4.9. For instance, consider BAT#1 and BAT#2 in Fig. 1(a). If $\Delta V_{21} = V_{t2} - V_{t1}$ exceeds the predetermined voltage difference threshold value ΔV_{Th} , the controller commands the inductor current I_{L1} of BC#1 to be positive (in the same direction as indicated in Fig. 4.8) by setting the value of α to +1, facilitating charge transfer from BAT#2 to BAT#1. Conversely, if $\Delta V_{21} = V_{t2} - V_{t1}$ falls below $-\Delta V_{Th}$, the controller commands I_{L1} to be negative (opposite to its direction in Fig. 4.8) by setting α to -1, promoting charge transfer from BAT#1 to BAT#2. If the voltage difference lies within the range of $-\Delta V_{Th}$ and ΔV_{Th} , α maintains its current value (as indicated by the dashed range) until the voltage difference reaches either ΔV_{Th} or $-\Delta V_{Th}$. Although the BCs can remain operational continuously, they can optionally be deactivated once the balanced condition is attained to conserve power loss (conduction and switching power losses).

The balanced state is achieved when all voltage differences between battery cells in the pack are simultaneously within the range of $-\Delta V_{Th}$ and ΔV_{Th} . The enable/disable logic is presented in the upper section of Fig. 4.9, where ΔV_{ij} represents a general expression for all voltage differences in the pack. In a three-cell system, the voltage differences are ΔV_{21} , ΔV_{32} , and ΔV_{31} . The output of the enable/disable logic is the parameter λ , with a value of 1 to enable balancing operation and a value of 0 to disable it. As elaborated later, the presented balancing controller achieves open-circuit voltage balancing, offering a more precise representation of SOC compared to terminal voltage.

Nevertheless, users can opt to utilize the controller for terminal voltage balancing. During charging operation, the terminal voltage of a battery cell surpasses its open-circuit voltage, while during discharging, it falls below it due to the voltage drop across the battery's internal resistance R_s . Consequently, terminal voltage balancing may lead to overbalancing or underbalancing.

4.3.8. Simulation Results

The simulation results of the proposed inductor-based cell balancing system, employing a buck-boost topology, reveal a notable trend: the voltages of the three cells gradually converge. This convergence occurs as a result of the cell balancing process, wherein excess charge from higher voltage cells is transferred to lower voltage cells, thereby equalizing their voltages.

Over time, it becomes apparent that the voltages of the individual cells approach a merged state, reaching a point where all three voltages align closely with each other. Ultimately, this alignment results in the merged voltage level becoming equivalent to the output voltage of the system.

This observed phenomenon underscores the effectiveness of the inductor-based cell balancing approach in achieving voltage equalization among the battery cells. Additionally, it highlights the importance of proper voltage management in maximizing the performance and longevity of battery packs.

Moreover, the convergence of cell voltages indicates successful balancing operation, wherein the system effectively redistributes charge to ensure uniformity across all cells. This not only enhances overall battery pack efficiency but also contributes to improved safety and reliability in battery-powered applications.

In conclusion, the simulation results validate the efficacy of the proposed inductor-based cell balancing system, emphasizing its capability to harmonize cell voltages and optimize battery performance within a buck-boost topology framework.

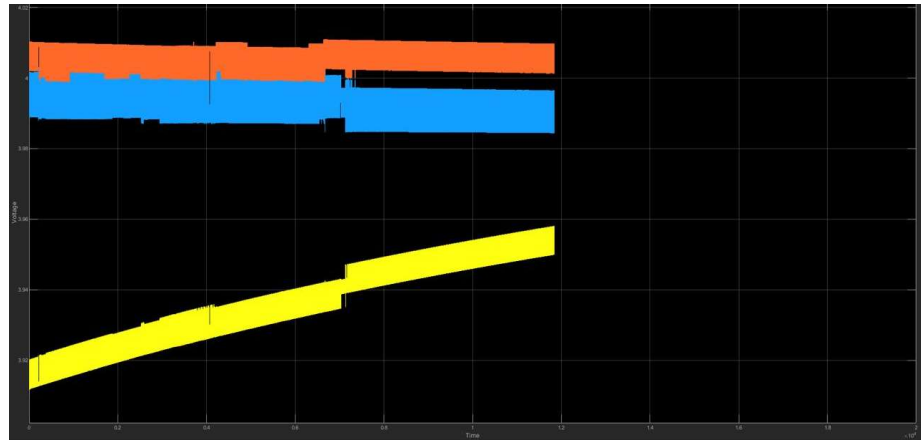


Figure 4.20: Voltage Output of the three Cells

5. Project Outcomes

Implementing a Battery Management System (BMS) for a microgrid, incorporating cell balancing and State of Charge (SoC) estimation, can yield several significant outcomes:

1. **Enhanced Battery Performance:** By continuously monitoring and balancing the cells within the battery pack, the BMS ensures that each cell operates within its optimal range. This optimization leads to increased battery lifespan and improved overall performance.
2. **Improved Energy Efficiency:** Accurate SoC estimation allows for better management of energy storage, ensuring that the battery is neither overcharged nor over-discharged. This optimization helps maximize the efficiency of energy storage and utilization within the microgrid.
3. **Enhanced Reliability and Safety:** Cell balancing helps mitigate issues such as cell degradation, capacity mismatches, and thermal runaway, thereby enhancing the safety and reliability of the battery system. This is crucial for ensuring uninterrupted power supply within the microgrid.
4. **Optimized Microgrid Operation:** A well-implemented BMS enables the microgrid to adapt to changing energy demands and sources efficiently. By accurately estimating SoC and managing battery health, the BMS contributes to the overall stability and resilience of the microgrid, supporting its seamless integration with renewable energy sources and improving its ability to handle fluctuations in energy supply and demand.
5. **Data Insights for Maintenance and Optimization:** The BMS continuously collects data on battery performance, including cell voltages, currents, temperatures, and SoC. Analyzing this data provides valuable insights into battery health, usage patterns, and potential optimization opportunities. This data-driven approach facilitates proactive maintenance, early fault detection, and continuous improvement of the microgrid's energy storage system.

Overall, the successful implementation of a Battery Management System with cell balancing and SoC estimation capabilities can significantly enhance the performance, reliability, and cost-effectiveness of energy storage systems within microgrids, ultimately contributing to the advancement of sustainable and resilient energy infrastructure.

References

- [1] M. T. Lawder et al., "Battery Energy Storage System (BESS) and Battery Management System (BMS) for Grid-Scale Applications," in Proceedings of the IEEE, vol. 102, no. 6, pp. 1014-1030, June 2014, doi: 10.1109/JPROC.2014.2317451.
- [2] K. G. H. Mangunkusumo, K. L. Lian, P. Aditya, Y. . -R. Chang, Y. D. Lee and Y. H. Ho, "A battery management system for a small microgrid system," 2014 International Conference on Intelligent Green Building and Smart Grid (IGBSG), Taipei, Taiwan, 2014, pp. 1-6, doi: 10.1109/IGBSG.2014.6835277.
- [3] http://mit.edu/evt/summary_battery_specifications.pdf
- [4] Biswas, Md Multan & Azim, Md & Saha, Tonmoy Kumar & Zobayer, Umama & Urmi, Monalisa. (2013). Towards Implementation of Smart Grid: An Updated Review on Electrical Energy Storage Systems. Smart Grid and Renewable Energy. 4. 122-132. 10.4236/sgre.2013.41015.
- [5] Zhou, L.; Lai, X.; Li, B.; Yao, Y.; Yuan, M.; Weng, J.; Zheng, Y. State Estimation Models of Lithium-Ion Batteries for Battery Management System: Status, Challenges, and Future Trends. Batteries 2023, 9, 131.
- [6] R, Rakshitha. (2020). Integration of Coulomb Counting Method in Battery Management System for Electric Vehicle. International Journal of Engineering Research and. V9. 10.17577/IJERTV9IS050639.
- [7] Ng, Kong & Moo, Chin-Sien & Chen, Yi-Ping & Hsieh, Yao-Ching. (2009). Enhanced Coulomb counting method for estimating state-of-charge and state-of-health of lithium-ion batteries. Applied Energy. 86. 1506-1511. 10.1016/j.apenergy.2008.11.021.
- [8] M. Lagraoui, S. Doubabi and A. Rachid, "SOC estimation of Lithium-ion battery using Kalman filter and Luenberger observer: A comparative study," 2014 International Renewable and Sustainable Energy Conference (IRSEC), Ouarzazate, Morocco, 2014, pp. 636-641, doi: 10.1109/IRSEC.2014.7059849.
- [9] P. S. Babu and K. Ilango, "Comparative Analysis of Passive and Active Cell Balancing of Li Ion Batteries," 2022 Third International Conference on Intelligent Computing Instrumentation and Control Technologies (ICICICT), Kannur, India, 2022, pp. 711-716, doi: 10.1109/ICICICT54557.2022.9917778.
- [10] M. K. Al-Smadi and J. A. Abu Qahouq, "Evaluation of Current-Mode Controller for Active Battery Cells Balancing With Peak Efficiency Operation," in IEEE Transactions on Power Electronics, vol. 38, no. 2, pp. 1610-1621, Feb. 2023, doi: 10.1109/TPEL.2022.3211905.

- [11] A. T. Elsayed, C. R. Lashway and O. A. Mohammed, "Advanced Battery Management and Diagnostic System for Smart Grid Infrastructure," in *IEEE Transactions on Smart Grid*, vol. 7, no. 2, pp. 897-905, March 2016, doi: 10.1109/TSG.2015.2418677.
- [12] R. Abdallah, A. -L. Gehin and J. -Y. Dieulot, "Battery Lifetime Optimization in a Solar Microgrid," 2022 IEEE 17th International Conference on Control & Automation (ICCA), Naples, Italy, 2022, pp. 154-159, doi: 10.1109/ICCA54724.2022.9831924
- [13] K. Prompinit and S. Khomfoi, "Battery Management System for Microgrid Applications," 2018 15th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON), Chiang Rai, Thailand, 2018, pp. 245-248, doi: 10.1109/ECTICon.2018.8619879.
- [14] J. Garche and A. Jossen, "Battery management systems (BMS) for increasing battery life time," *TELESCON 2000. Third International Telecommunications Energy Special Conference (IEEE Cat. No.00EX424)*, Dresden, Germany, 2000, pp. 81-88, doi: 10.1109/TELESC.2000.918409.
- [15] B. Devi and V. S. Kumar, "Lithium-ion Battery Management System: A review," 2022 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2022, pp. 1-6, doi: 10.1109/ICPECTS56089.2022.10047494.
- [16] R. Hiremath, S. Hulakund, V. R. Torgal, A. Patil, H. R. Patil and A. B. Raju, "State-Of-Charge Estimation using Extended Kalman Filter," 2023 International Conference for Advancement in Technology (ICONAT), Goa, India, 2023, pp. 1-6, doi: 10.1109/ICONAT57137.2023.10080761.
- [17] R, Rakshitha. (2020). Integration of Coulomb Counting Method in Battery Management System for Electric Vehicle. *International Journal of Engineering Research and*. V9. 10.17577/IJERTV9IS050639.
- [18] S. S. Shiekh, K. L. Khan, T. N. Mir and A. H. Bhat, "Analysis of Extended Kalman Filter for SOC Estimation in Lithium-ion Batteries," 2022 Second International Conference on Power, Control and Computing Technologies (ICPC2T), Raipur, India, 2022, pp. 1-6, doi: 10.1109/ICPC2T53885.2022.9776928.
- [19] Kong Soon Ng, Chin-Sien Moo, Yi-Ping Chen, Yao-Ching Hsieh, Enhanced coulomb counting method for estimating state-of-charge and state-of-health of lithium-ion batteries, *Applied Energy*, Volume 86, Issue 9, 2009, Pages 1506-1511, ISSN 0306-2619,